

Trainings ▾

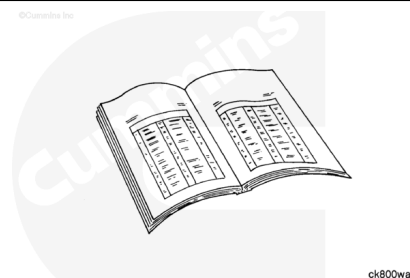
Preparatory Steps

Remove the fan pulley. Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/100/100-008-039.html>).

Remove the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>).

Remove the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>).

Remove the camshaft. For C Series diesel engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>) in Troubleshooting and Repair Manual, C Series Engines, Bulletin 3666003. For C Series gas engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/39/39-001-008.html>) in Troubleshooting and Repair Manual, C8.3G (Natural Gas) Engines, Bulletin 3666206.

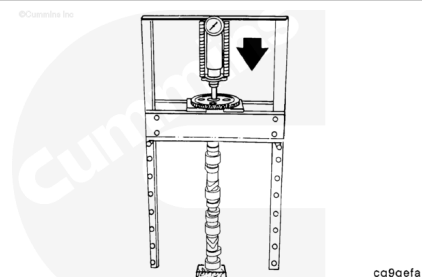


Remove

⚠ CAUTION ⚠

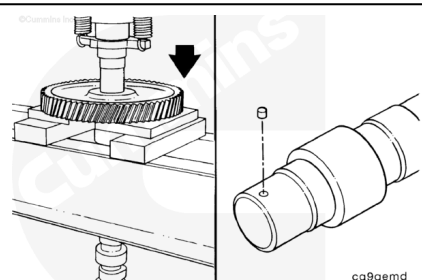
Place a wooden block under the camshaft to avoid damage as the camshaft drops free from the cam gear.

Place the camshaft and gear in a hydraulic press. Press the gear off the camshaft.



Remove the camshaft dowel pin.

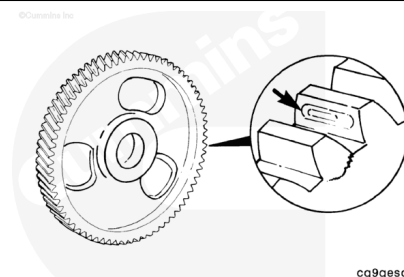
Remove burrs with Scotch-Brite 7448, Part No. 3823258.



Inspect for Reuse

Inspect the camshaft gear for cracked, chipped, or broken teeth.

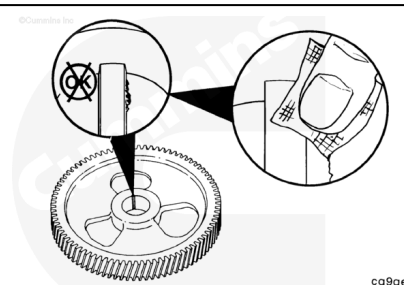
Note : If the fretting, burrs, or raised material can not be removed with Scotch-Brite 7448, Part No. 3823258, replace the camshaft gear.



Inspect the camshaft gear bore for fretting or burrs and the keyway for burrs.

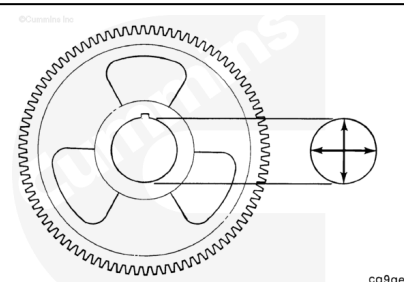
Remove burrs with Scotch-Brite 7448, Part No. 3823258.

Note : If the keyway is damaged or the burrs can not be removed, the camshaft gear must be replaced.



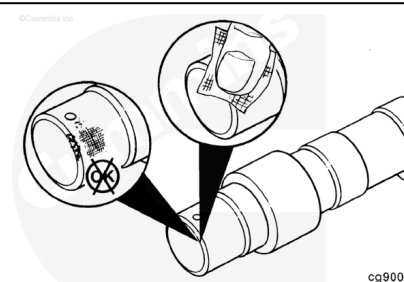
Measure the camshaft gear bore inside diameter.

mm		in
41.48	MIN	1.633
41.51	MAX	1.634



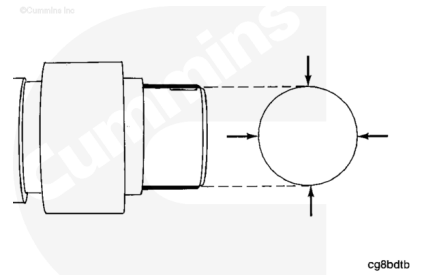
Inspect the camshaft nose for fretting or burrs.

Note : If fretting or burrs can not be removed with Scotch-Brite 7448, Part No. 3823258, the camshaft must be replaced.



Measure the camshaft gear journal outside diameter.

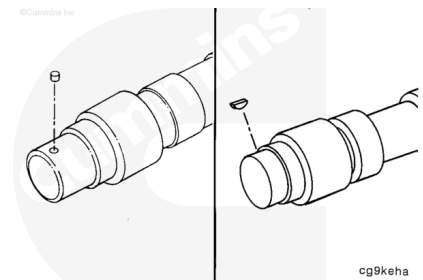
mm		in
41.56	MIN	1.636
41.58	MAX	1.637



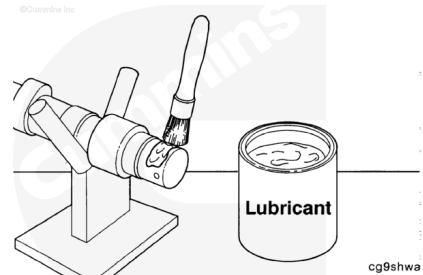
Install

Use a leather hammer to install the Woodruff key or dowel pin.

Note : The steel camshaft, Part No. 3923388, uses a dowel pin. The cast-iron camshaft, Part No. 3924471, uses a Woodruff key.



Lubricate the camshaft surface with clean engine oil.

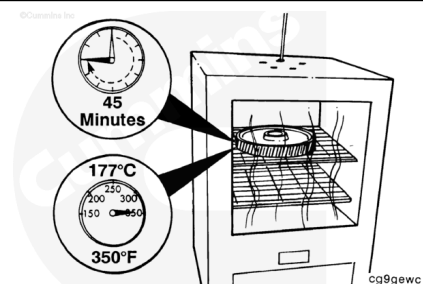


⚠ WARNING ⚠

Wear protective gloves to prevent personal injury when handling parts that have been heated.

Steel Camshaft Gear

Heat the camshaft gear in a preheated oven for a minimum of 45 minutes; do **not** exceed 6 hours.

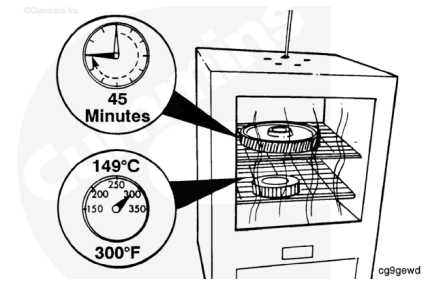


Measurements

	celsius	fahrenheit
Temperature	177	350

⚠ CAUTION ⚠

The gear will be permanently distorted if overheated. The oven temperature must never exceed the specification.



Tempered Camshaft Gear

Note : The steel camshaft, Part No. 3918777, uses a dowel pin. The cast-iron camshaft, Part No. 3924471, uses a Woodruff key.

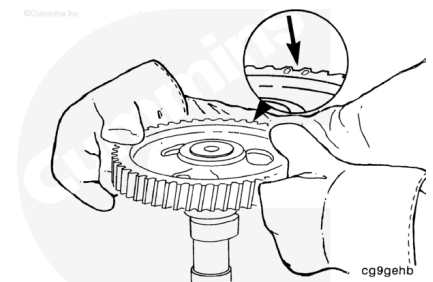
Measurements

	celsius	fahrenheit
Temperature	149	300

⚠ WARNING ⚠

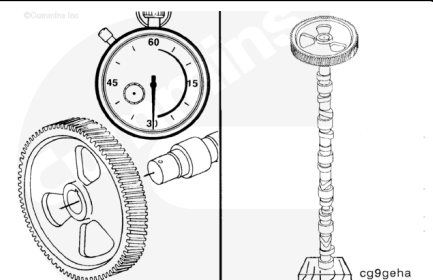
Wear protective gloves to prevent personal injury when handling parts that have been heated.

Install the camshaft gear with the timing marks away from the camshaft.



⚠ CAUTION ⚠

To prevent engine damage, the timing marks and gear part number must be facing away from the camshaft when the gear is installed. Keep the camshaft in a vertical position with the gear up until the gear has cooled. Do not use water to reduce the cooling time. Using water will crack the gear.



Note : The gear must be installed within 30 seconds after it is removed from the oven.

Remove the gear from the oven. Align the keyway in the gear with the dowel pin in the camshaft; install the gear on the camshaft. Make sure the gear is seated against the camshaft shoulder.

Finishing Steps

Install the camshaft. For C Series diesel engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>) in Troubleshooting and Repair Manual, C Series Engines, Bulletin 3666003. For C Series natural gas engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/39/39-001-008.html>) in Troubleshooting and Repair Manual, C8.3G (Natural Gas) Engines, Bulletin 3666206.



Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>).

Install the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>).

Install the fan pulley. Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/100/100-008-039.html>).

Operate engine and check for leaks.